

SUJAY PATIL

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Education

University of Glasgow

Integrated Master of Science (MSci) in Computer Science

Sep 2021 – July 2026

Glasgow, Scotland, UK

Experience

Glasgow University Software Service

Software Engineer

Mar 2024 – Mar 2026

Glasgow, Scotland, UK

- Translated Analogue Circuit Design Tuning MatLab Codes into Production Python Code using **ML algorithms**.
- Implemented **Machine Learning Techniques** to build and train **neural networks** for **constraint optimization**.
- Deployed **licensed proprietary software** compatible across multiple **Linux** distributions via a custom GUI installer.
- Built **test cases** to guarantee the optimisation algorithm's accuracy and efficiency.
- Technologies used: **NumPy, PyTorch, MATLAB**

Barclays

Technology Analyst Intern

Jun 2025 – Aug 2025

Glasgow, Scotland, UK

- Operated as a **Risk & Control Partner** within the **Markets Pre-Trade Technology Team**.
- Created a comprehensive list of annual **supplier contractual obligation waivers** across 5 different risk horizontals.
- Reduced outstanding **sub-limited JIRA tickets** by approximately **40%**, improving the team's risk remediation.
- Identified **40+ Cyber waiver tickets** to raise them to limited risk status in **ORAC** to enable tracking and oversight.
- Automated a **risk ownership mapping** process, reducing manual reporting effort by **2–3 hours per week**.
- Technologies used: **Atlassian Jira, ORAC, Microsoft Excel, Microsoft Teams**

National Manufacturing Institute Scotland

Software Engineer Intern

Jun 2024 – Jun 2025

Paisley, Scotland, UK

- Developed an **AI system** that can read, extract, and classify invoice data to carry out complete carbon accounting.
- Integrated all **Scope 3 emission categories** based on the GHG protocol, alongside full carbon accounting functions.
- Built a **Machine Data Analysis** system for **80+ machines** with real-time energy insights and automated reporting.
- Developed an **interactive dashboards** to highlight load imbalance, idle consumption, and batch-level cost metrics.
- Technologies used: **Python, Django, React.js, SQL, OpeanAI API, Figma, Git, Microsoft Excel**

GU Orbit

Head Software Engineer

Sep 2022 – Jun 2025

Glasgow, Scotland, UK

- Developed the prototype of the **processing pipeline** to be deployed on an **aerial testing platform**.
- Developed Python and C++ scripts and interfaced **BMP180 sensors** to **Raspberry Pi** with **i2c connection**.
- Successfully completed the **CloudView balloon launch**. The trip lasted 3 hours and sent back large amounts of data.
- Technologies used: **C++, Python, Docker, Raspberry Pi, Git, Microsoft Teams**

Projects

AR-Assisted Sign Language Learning | *PyTorch, React.js, MediaPipe, ONNX Runtime Web*

Sep 2025 - Apr 2026

- For my Master's project, I researched **AR-assisted sign language education** by developing a browser based **BSL learning platform** with real-time gesture recognition and feedback. The system combined MediaPipe hand tracking, neural networks, and DTW based motion analysis to evaluate static and dynamic gestures, and was evaluated through a user study on engagement, usability, and accessibility.

AI-Powered Life Cycle Assessment Tool - LCAi | *Django REST, React.js, OpenAI API*

Sep 2024 - Mar 2025

- In collaboration with the National Manufacturing Institute Scotland, I developed an AI-powered Lifecycle Assessment (LCA) tool designed to evaluate the **environmental impact** of products. The tool automatically analyzes **Bills of Materials (BOMs)** and related product data to assess environmental impacts across the product's entire lifecycle.

UofG Gaming Lab - Project Gundam | *Unity, C#, Google Cloud*

Sep 2023 - Mar 2024

- We developed a **Mobile AR application** for the **WorldCon 2024** event, using Unity. To encourage attendees, to explore Glasgow, this app aims to enhance visitors' experiences of the city. In addition the app also provides the user with a compass and a Gundam tracker, The app spawns several sorts of Gundam Models across Glasgow.

Technical Profile

Languages: Python, Java, C, C++, C#, HTML, CSS, JavaScript, TypeScript, SQL, MATLAB

Frameworks: Django REST, Flask, React.js, NumPy, Unity, PyTorch, Robot Operating System (ROS)

Technical skills: Git, Full Stack Development, Artificial Intelligence, Project Management, Business Analytics, LaTeX, Risk Analysis, Data Engineering, Data Analysis, Cloud Computing, Machine Learning, Deep Learning Algorithms, Data Structures, Database Modeling, Big Data Systems, Systems Programming, Solutions Engineering